

Data-Adaptive Variable Curvature for Hyperbolic Embeddings of Medical Ontologies

A Research Proposal

Abstract

Hyperbolic embeddings have been established as the natural geometry for tree-structured medical ontologies. Recent work has shown that variable-curvature manifolds further improve over constant-curvature hyperbolic embeddings for heterogeneous hierarchies, either through intrinsic geometric variation (complex hyperbolic, Xiao et al. 2021) or through radially-symmetric variation in warped products (Di Giovanni et al. 2022). However, in all existing approaches, curvature variation is determined by intrinsic geometric properties of the manifold, not by data-dependent properties of the ontology being embedded. We propose the first method in which curvature is explicitly derived from ontology-level structural statistics. Specifically, we anchor curvature to node-level branching factor and subtree size, smoothed into a continuous curvature field via kernel weighting. We empirically motivate the approach by characterizing structural heterogeneity in ICD-10-CM: across 255 sections with at least three internal nodes, mean branching factor varies 4.29-fold (from 2.33 to 10.00), and per-chapter subtree size varies 10-fold. We propose two complementary algorithmic instantiations: (i) tree-anchored curvature, where node-level κ values are computed once and used throughout training, and (ii) alternating optimization, where curvature is iteratively refined from smoothed embedding density. We address geodesic-distance tractability via line-integral approximation with explicit error bounds. Downstream evaluation targets rare-disease retrieval with strict subtree holdout. We compare against six baselines spanning Euclidean, constant-curvature Poincaré, learnable global curvature, mixed-curvature product spaces, complex hyperbolic, and heterogeneous radial-curvature manifolds. Phase 1 (data pipeline) is complete; the proposal outlines remaining phases.

1. Introduction and Motivation

Rare-disease misdiagnosis is a documented clinical crisis. Rare-disease patients average roughly seven specialist visits over five or more years before receiving a correct diagnosis. AI-augmented decision support struggles to surface adjacent diagnoses — diseases that branch off one or two steps away in the medical taxonomy — leading to systematic misses for rare conditions whose symptom patterns overlap with more common diseases.

Modern medical AI systems represent disease concepts as dense vectors in flat Euclidean embedding spaces. However, medical ontologies are fundamentally hierarchical, and trees cannot be embedded in Euclidean space without distortion that grows logarithmically in the size of the tree (Linial et al., 1995). Hyperbolic geometry, with its exponential volume growth, is the natural setting for tree-structured data. Nickel and Kiela (2017) and Beam et al. (2018) demonstrated this for general taxonomies and medical concepts respectively. Recent work (Yang et al., 2025; Kim et

al., 2026) has extended hyperbolic methods to additional medical retrieval and reasoning tasks.

Variable-curvature manifolds have been shown to outperform constant-curvature hyperbolic embeddings on heterogeneous hierarchies. Xiao et al. (2021) introduced complex hyperbolic embeddings via the unit-ball Bergman-metric model, with sectional curvature pinched between -1 and $-1/4$ depending on local direction; they showed improvements on ICD-10 and similar taxonomies. Di Giovanni et al. (2022) proposed heterogeneous rotationally-symmetric manifolds via warped products, allowing radial curvature variation while preserving closed-form pairwise distances. Both approaches confirm that constant curvature is suboptimal for real-world hierarchical data.

However, in both Xiao et al. and Di Giovanni et al., curvature variation is determined by intrinsic geometric properties of the chosen manifold — the Bergman metric in the complex case, rotational symmetry in the warped-product case. Curvature is fixed by the geometric structure rather than learned from data. This means curvature does not adapt to where in the ontology structural heterogeneity actually concentrates. In real medical ontologies, different chapters, sections, and subtrees have very different branching and depth-density properties; an ideal embedding geometry would have curvature that varies in response to where the data actually demands it.

We propose the first method in which curvature is data-adaptive: explicitly derived from node-level structural statistics of the ontology being embedded. Local branching factor and subtree size define each node's preferred curvature; these discrete values are smoothed into a continuous $\kappa(x)$ field that defines the embedding manifold's Riemannian metric. The result is a geometry that fits the ontology's local structure rather than imposing a fixed geometric pattern on it.

2. Background

2.1 Constant-Curvature Hyperbolic Embeddings

The Poincaré ball model represents hyperbolic space as the open unit ball with metric $g_x = (4 / (1 - ||x||^2)^2) \cdot g^E$. The induced distance is:

$$d(u, v) = \text{arcosh}(1 + 2||u - v||^2 / [(1 - ||u||^2)(1 - ||v||^2)])$$

Nickel and Kiela's training procedure uses Riemannian stochastic gradient descent with the negative-sampling loss $L(\Theta) = \sum \log[\exp(-d(u,v)) / \sum \exp(-d(u,v'))]$ and update rule $\theta_{t+1} \leftarrow \text{proj}(\theta_t - \eta \cdot ((1 - ||\theta_t||^2)^2/4) \cdot \nabla^E L)$.

2.2 Existing Variable-Curvature Approaches

Complex hyperbolic (Xiao et al. 2021). Uses the unit ball in d complex dimensions with the Bergman metric, which has variable sectional curvature pinched between -1 (in complex projective directions) and $-1/4$ (in totally real directions). Closed-form distance: $d(u,v) = \text{arcosh}(|\langle u,v \rangle| / \sqrt{(1-|u|^2)(1-|v|^2)})$, where $\langle \cdot, \cdot \rangle$ is a Hermitian form. Curvature variation is determined by the complex geometric structure of the manifold, not by the data being embedded.

Heterogeneous radial-curvature manifolds (Di Giovanni et al. 2022). Uses warped product manifolds of the form $M_h \times_{\lambda^2} \mathbb{R}$, where a homogeneous factor is combined with a radial coordinate weighted by a learnable function $\lambda(r)$. Closed-form distances are retained. Curvature varies along the radial direction only; the angular directions remain homogeneous. Variation is data-aware in that $\lambda(r)$ is learned, but it is constrained to depend only on radial position.

Mixed-curvature product spaces (Gu et al. 2019). Cartesian product of multiple constant-curvature factor spaces, e.g., $\mathbb{H} \times \mathbb{S} \times \mathbb{R}$. Each factor has its own fixed or learned curvature; within a factor, curvature is uniform. Curvature varies across factors, not across positions within the manifold.

These methods establish that variable curvature helps; they leave open the question of whether tying curvature directly to data structure (rather than to geometric structure) yields further improvements.

3. Empirical Motivation: Heterogeneity in ICD-10-CM

We parsed the FY2025 ICD-10-CM tabular XML from CMS into a unified tree with one synthetic root, 22 chapters, 296 sections, and 46,498 diagnosis nodes (46,817 total nodes, 46,816 edges). Of these, 10,775 are internal nodes (branching factor ≥ 1).

Chapter-level mean branching factor varies modestly (1.45-fold ratio: 3.67 to 5.32) with within-chapter standard deviations of 1.3 to 2.8. Section-level analysis reveals substantially larger heterogeneity: across 255 sections with at least three internal diagnosis nodes, mean branching factor ranges from 2.33 to 10.00, a 4.29-fold ratio. Maximum subtree size per chapter ranges from 37 (Chapter 1) to 361 (Chapter 19), an approximately 10-fold ratio. The distribution of branching factors has a long right tail: while most internal nodes have branching factor 2-5, rare nodes reach 20-34. These statistical features motivate a curvature scheme that depends explicitly on local structural properties, not merely on intrinsic manifold geometry.

4. Method

4.1 Per-Node Curvature from Tree Statistics

For each internal node v , we compute branching factor $b(v)$ and subtree size $s(v)$ directly from the ICD-10-CM tree. We motivate the dependence on b through a volumetric argument: in d -dimensional hyperbolic space of curvature K , the volume at radius r grows as $\exp((d-1)r\sqrt{|K|})$, while the number of leaves at depth $\square$ in a tree of branching factor b is $b^\square$. Matching these, under a depth-radius correspondence, yields a leading-order target $\sqrt{|K|} \propto \log(b)$. We retain the volumetric motivation but treat the precise functional form as an empirically validated parametric family. Specifically, we define a structural feature $\rho(v)$ and an associated per-node curvature:

$$\begin{aligned}\rho(v) &= \alpha \cdot \log(b(v))^2 + \beta \cdot \log(s(v)) \\ \kappa(v) &= \kappa_0 \cdot (1 + \gamma \cdot \rho(v))\end{aligned}$$

with $\kappa_0 < 0$ a baseline curvature and $\alpha, \beta, \gamma \geq 0$ learned scalars subject to constraints ensuring $\kappa(v) < 0$ for all relevant ranges. The $\log(b)^2$ term aligns the parametric form with the volumetric derivation; the $\log(s)$ term captures subtree-density correction. We perform sensitivity analysis on (α, β, γ) and an ablation with $\gamma \rightarrow 0$ (recovering the constant-curvature baseline) to isolate the contribution of curvature variation. The Cartan-Hadamard preservation requirement reduces to verifying $\kappa(v) \leq 0$ across the parameter family; we constrain (α, β, γ) accordingly and report the verified safe region.

4.2 From Discrete to Continuous Curvature

The Riemannian metric must be defined at every point in the manifold, not just at the discrete embedded node positions. We construct a continuous curvature field $\kappa(x)$ by Gaussian kernel smoothing of the discrete node-level values:

$$\kappa(x) = \sum_v K_\sigma(d(x, f(v))) \cdot \kappa(v) / \sum_v K_\sigma(d(x, f(v)))$$

with $K_\sigma(d) = \exp(-d^2 / (2\sigma^2))$ and σ a learnable bandwidth parameter. At each embedded node position $f(v)$, the kernel weight on v dominates, so $\kappa(f(v)) \approx \kappa(v)$; between nodes, the field interpolates smoothly. The conformal factor $\lambda(x)$ is derived from $\kappa(x)$ via the standard relation for two-dimensional conformal metrics and extended to higher dimensions using the formula $K = -\lambda^{-2} \Delta \log \lambda$ for conformally Euclidean metrics, with $\lambda(x)$ recovered by numerical solution on a discretized grid when no closed form exists.

4.3 Two Algorithmic Instantiations

Tree-anchored curvature. Compute $\rho(v)$ and $\kappa(v)$ once from tree statistics. During training, the curvature field $\kappa(x)$ is reconstructed at each step from current embedding positions via kernel smoothing of fixed node-level values. The structural anchors $\kappa(v)$ never change; only embeddings evolve. This formulation completely decouples curvature from embedding density and removes any circular dependency.

Alternating optimization. Initialize as in the tree-anchored variant. After every K epochs, recompute a smoothed curvature field $\tilde{\kappa}(x)$ by kernel-weighting current embedding positions rather than fixed node values, then continue training. Alternate between embedding updates and curvature refinement until convergence. To maintain stability across changing curvature, we use a curvature-aware Riemannian optimizer (Bécigneul and Ganea 2019; recent variants in Wang et al. 2024) that adjusts retraction based on current local geometry. We track embedding norm stability and report empirical convergence behavior; if convergence is not observed, we report performance at fixed iteration counts.

4.4 Tractable Distance Computation with Error Bounds

For our variable-curvature conformally Euclidean manifold, geodesic distance lacks a closed form. Numerical ODE integration is feasible but expensive at training scale ($\sim 10^8$ distance computations per epoch). We approximate geodesic distance by the line integral along the straight Euclidean segment between embedded points:

$$d(u, v) = \int_0^1 \|u - v\| / \lambda(u + t(v - u)) dt$$

evaluated by Gauss-Legendre quadrature. We provide an a priori error bound:

$$|d(u,v) - \hat{d}(u,v)| \leq C \cdot \|u - v\|^2 \cdot \sup |\nabla \log \lambda|$$

where C depends on dimension and the Lipschitz constant of the metric. Since λ in our parametric family is bounded and smooth, $\sup |\nabla \log \lambda|$ is bounded. We additionally verify the approximation empirically: a held-out subset of pairs ($\sim 1,000$ per training run) has its geodesic distance computed exactly via Runge-Kutta ODE integration and compared to the line-integral approximation. We report mean and worst-case approximation error and fall back to ODE-based integration if the worst-case exceeds a preset threshold.

5. Experimental Plan

5.1 Phase 1: Data Pipeline (complete)

ICD-10-CM tabular XML parsed into a unified tree with full structural statistics. Per-node features (b, s, depth, chapter, section) computed and persisted.

5.2 Phase 2: Baselines

Six baseline methods, each trained and evaluated under identical protocol: (i) Euclidean embedding with cosine similarity; (ii) constant-curvature Poincaré ball, $K = -1$; (iii) learnable global curvature K trained as a single scalar parameter; (iv) mixed-curvature product spaces (Gu et al. 2019), with factor curvatures learned; (v) complex hyperbolic embeddings in the unit-ball Bergman-metric model (Xiao et al. 2021), using authors' released code adapted to our ICD-10-CM tree; (vi) heterogeneous radial-curvature manifolds (Di Giovanni et al. 2022) with learnable radial curvature function. We omit hyperbolic entailment cones (Ganea et al. 2018), as they address asymmetric is-a relations — orthogonal to our heterogeneity axis. We additionally report constructive embeddings (HS-DTE; Sala et al. 2018) as a reconstruction upper bound, recognizing they are not learned and do not generalize to downstream tasks.

5.3 Phase 3: Novel Methods and Ablations

Both algorithmic instantiations (tree-anchored, alternating) are trained with the same loss and evaluation protocol as baselines, at embedding dimensions 5, 10, and 50. We perform a γ -sweep across $\{0, 0.25, 0.5, 1.0, 2.0\}$ to isolate the effect of curvature variation, with $\gamma = 0$ recovering the constant-curvature baseline. We additionally report curvature-matching diagnostics: the average per-node distortion between learned $\kappa(v)$ and discrete Forman curvature computed directly on the ICD-10-CM graph, as a direct test of the central hypothesis that local curvature should track local structural properties.

5.4 Phase 4: Rare-Disease Retrieval

We jointly embed ICD-10-CM, HPO phenotypes, and Mondo disease concepts into the same manifold. HPO terms are added as new nodes with edges to their associated diseases via HPOA annotations and to their hierarchical parents in HPO. Joint embedding allows geometric scoring between a query symptom set and candidate diseases via summed hyperbolic distances. To prevent leakage, we hold out entire

Orphanet sub-branches: all test diseases and their HPO annotations are excluded from training data. The model must retrieve held-out diseases from their characteristic symptoms at evaluation time. Metrics: top-k recall ($k \in \{1, 5, 10, 50\}$), mean reciprocal rank, reported overall and stratified by disease rarity, local branching density at the disease's ontology position, and tree depth.

5.5 Compute

All training on a single A100 GPU (Google Colab Pro+). Estimated total: ~300 A100-hours across all methods and dimensions, including line-integral approximation for variable-curvature methods and exact distances for baselines.

6. Expected Contributions

(C1) The first empirical characterization of structural heterogeneity in ICD-10-CM at section-level granularity, with multi-statistic analysis (branching factor and subtree size). (C2) The first method tying hyperbolic-embedding curvature to data-derived ontology statistics rather than to intrinsic manifold geometry. (C3) Explicit construction of a continuous curvature field via kernel smoothing of discrete node-level values, with the inverse problem of recovering $\lambda(x)$ from $\kappa(x)$. (C4) Empirical verification of line-integral distance approximation against numerical geodesic integration with a priori error bounds. (C5) Comparative evaluation of two algorithmic instantiations against six baselines spanning Euclidean, constant-curvature hyperbolic, learnable global curvature, mixed-curvature product spaces, complex hyperbolic, and heterogeneous radial-curvature manifolds — with γ -sweep ablation isolating the role of curvature variation. (C6) Rare-disease retrieval evaluation with strict subtree holdout to prevent cross-ontology leakage. (C7) Open-source release of code, trained embeddings, and parsed ontology data.

7. Risks and Mitigations

Differentiation from existing variable-curvature methods. Both complex hyperbolic and heterogeneous radial-curvature methods already achieve variable curvature. Our contribution is the first to derive curvature from data, not geometry. We will report head-to-head comparisons on identical protocol and characterize where data-adaptive curvature provides additional improvement over geometry-fixed alternatives. Should improvements not be robust, this is itself a meaningful empirical finding. **Cartan-Hadamard preservation.** We constrain the parametric family to a verified safe region. **Distance-approximation error.** Bounded analytically and verified empirically; ODE fallback available. **Alternating-optimization convergence.** Monitored empirically with curvature-aware optimization; if convergence is not observed, we report performance at fixed iteration counts. **Empirical generalization.** Per-chapter, per-section, and stratified reporting localizes where the method helps.

8. Timeline

Phase 1 (data pipeline): complete. Phase 2 (six baselines): weeks 1–8. Phase 3 (novel methods, math derivations, ablations): weeks 9–18. Phase 4 (rare-disease retrieval, joint

embedding, stratified evaluation): weeks 16-22. Writing and revision: weeks 22-26. Target submission: Regeneron Science Talent Search, mid-November 2026.

9. Conclusion

Variable-curvature hyperbolic embeddings have been established as superior to constant-curvature counterparts for heterogeneous hierarchical data, but in all existing methods the curvature variation is determined by intrinsic geometric properties of the chosen manifold rather than by data-dependent properties of the ontology being embedded. We propose the first method tying curvature directly to data-derived ontology statistics, with two complementary algorithmic instantiations, explicit kernel-based construction of a continuous curvature field, line-integral distance approximation with verified error bounds, and comparative evaluation against six baselines including the strongest existing variable-curvature methods. Phase 1 (empirical characterization of ICD-10-CM heterogeneity, data pipeline) is complete; subsequent phases implement and evaluate the proposed methods.

References

- Beam, A. L., Beaulieu-Jones, B. K., and Kohane, I. S. (2018). Learning Contextual Hierarchical Structure of Medical Concepts with Poincaré Embeddings to Clarify Phenotypes. *arXiv:1811.01294*.
- Bécigneul, G., and Ganea, O.-E. (2019). Riemannian Adaptive Optimization Methods. *International Conference on Learning Representations*.
- Di Giovanni, F., Luise, G., and Bronstein, M. M. (2022). Heterogeneous Manifolds for Curvature-Aware Graph Embedding. *arXiv:2202.01185*. *ICLR Workshop on Geometrical and Topological Representation Learning*.
- Ganea, O., Bécigneul, G., and Hofmann, T. (2018). Hyperbolic Entailment Cones for Learning Hierarchical Embeddings. *International Conference on Machine Learning*.
- Gu, A., Sala, F., Gunel, B., and Ré, C. (2019). Learning Mixed-Curvature Representations in Product Spaces. *International Conference on Learning Representations*.
- Kim, S., et al. (2026). GRail: Geometry-Aware Retrieval-Augmented Inference with LLMs over Hyperbolic Representations of Patient Trajectories. *arXiv:2602.12828*.
- Linial, N., London, E., and Rabinovich, Y. (1995). The Geometry of Graphs and Some of Its Algorithmic Applications. *Combinatorica*, 15(2), 215–245.
- Macaulay, M., Darling, A., and Fourment, M. (2023). Fidelity of Hyperbolic Space for Bayesian Phylogenetic Inference. *PLOS Computational Biology*, 19(4), e1011084.
- Nickel, M., and Kiela, D. (2017). Poincaré Embeddings for Learning Hierarchical Representations. *Advances in Neural Information Processing Systems* 30.
- Sala, F., De Sa, C., Gu, A., and Ré, C. (2018). Representation Tradeoffs for Hyperbolic Embeddings. *International Conference on Machine Learning*.
- Sarkar, R. (2011). Low Distortion Delaunay Embedding of Trees in Hyperbolic Plane. *Graph Drawing*.
- Wang, X., et al. (2024). Curvature-Aware Riemannian Optimization for Hyperbolic Learning. *arXiv:2405.13979*.
- Xiao, H., Liu, X., Song, Y., Wong, G. Y., and See, S. (2021). Unit Ball Model for Embedding Hierarchical Structures in the Complex Hyperbolic Space. *arXiv:2105.03966*. *NeurIPS submission*.
- Yang, X., et al. (2025). Hierarchical Retrieval with Out-Of-Vocabulary Queries: A Case Study on SNOMED CT. *arXiv:2511.16698*.